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SET 2 — Customer Feedback Analysis

According to your experiment sheet, Set 2 uses the **Customer Satisfaction** dataset and requires a histogram, pie chart, stacked bar chart, and word cloud.

Dataset

Customer ID	Age	Satisfaction Score
1	25	4
2	30	5
3	35	3
4	28	4
5	40	5

AIM

To analyze customer satisfaction data using **R programming** and visualize customer age and satisfaction score distributions using different charts.

ALGORITHM

1. Start RStudio.
2. Create the customer satisfaction dataset.
3. Store Customer ID, Age, and Satisfaction Score.
4. Create a **histogram** to visualize the distribution of customer ages.
5. Create a **pie chart** to show the distribution of satisfaction scores.
6. Divide customers into different age groups.
7. Create a **stacked bar chart** to visualize satisfaction scores by age group.
8. Create sample open-ended feedback data.
9. Generate a **word cloud** to identify common customer sentiments.
10. Add suitable titles and labels.
11. Display the results.
12. Stop.

R CODE

```
customer <- data.frame(  
  ID=c(1,2,3,4,5),  
  Age=c(25,30,35,28,40),  
  Satisfaction=c(4,5,3,4,5)  
)
```

1. Histogram

```
hist(customer$Age,  
      main="Distribution of Customer Ages",  
      xlab="Age",  
      ylab="Number of Customers")
```

2. Pie Chart

```
satisfaction <- table(customer$Satisfaction)  
  
pie(satisfaction,  
     main="Customer Satisfaction Scores",  
     labels=paste(names(satisfaction),satisfaction,sep=": "))
```

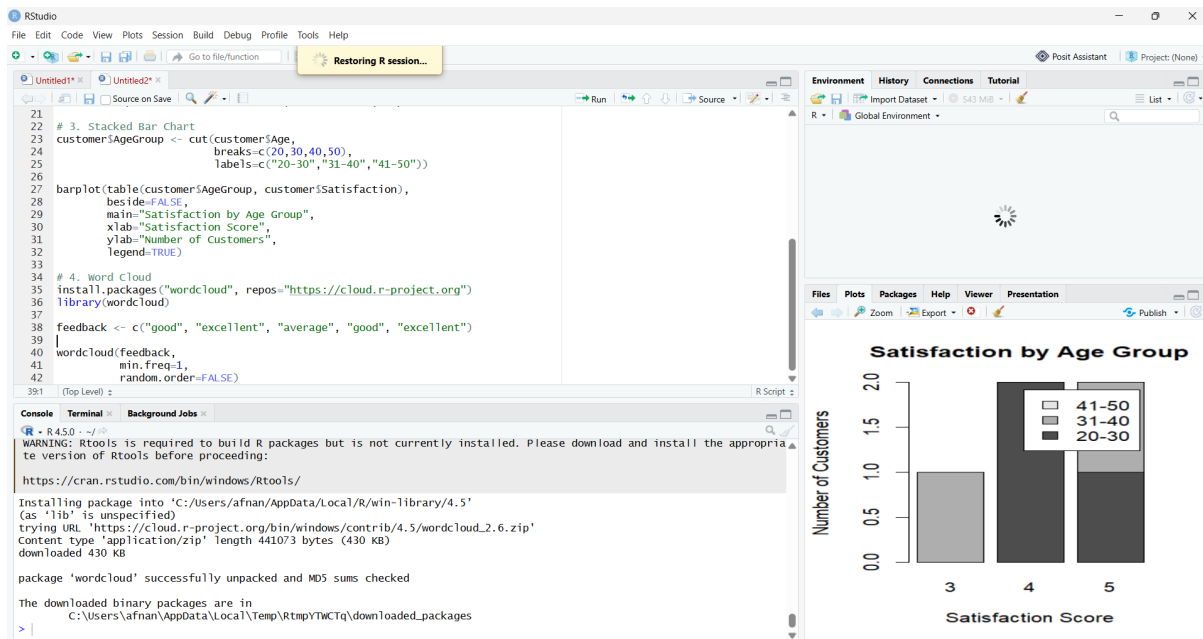
3. Stacked Bar Chart

```
customer$AgeGroup <- cut(customer$Age,  
                          breaks=c(20,30,40,50),  
                          labels=c("20-30","31-40","41-50"))  
  
barplot(table(customer$AgeGroup,customer$Satisfaction),  
        main="Satisfaction by Age Group",  
        xlab="Satisfaction Score",  
        ylab="Number of Customers",  
        legend=TRUE)
```

4. Word Cloud

```
install.packages("wordcloud",repos="https://cloud.r-project.org")  
library(wordcloud)  
  
feedback <- c("good","excellent","average","good","excellent")  
  
wordcloud(feedback,
```

```
min.freq=1,  
random.order=FALSE)
```



RESULT

Thus, the **Customer Feedback Analysis** was successfully performed using R programming. The histogram, pie chart, stacked bar chart, and word cloud were created to visualize customer age, satisfaction distribution, satisfaction by age group, and customer sentiments.